

Ramkumar Ramachandra

LLVM compiler engineer with over a decade of experience, and a track record in open source work, specializing in the middle-end. Hobby research includes constructing mathematical objects using a proof assistant.

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WORK

- Jan '26 – **Sr. Staff Compiler Engineer** Tenstorrent, United States
Present Upstream-first LLVM optimization. Fixed a hole in the constant-folder enabling folding of intrinsics. Made several improvements to vectorization, notably improving expansion of recipes, and SCEV expressions. Mentored an intern, guiding them to extend the optimization of cyclic-redundancy-check loops using carry-less multiply.
- Jun '24 – **Compiler Tech Lead** Codasip, Bristol, England
Jan '26 Upstream-first LLVM optimization for RISC-V: landed 400 patches spanning most of the middle-end. Authored a new analysis, HashRecognize, and enabled optimization of cyclic-redundancy-check loops using a table-lookup, with impact on several real-world programs including Linux. Introduced a carry-less multiply intrinsic with generic lowering. Significantly improved vectorization, notably authoring the CSE and constant-folder in VPlan. Drove the *icmp samesign* optimization effort.
- Mar '23 – **Senior Compiler Engineer** Imagination Technologies, Kings Langley, England
Feb '24 LLVM optimization for a RISC-V VPU. Made small improvements to several middle-end optimizations, with impact on embedded benchmarks. Introduced vector variants of the `lrint` intrinsic with custom lowering for RISC-V. Downstream work included improving the benchmarking infrastructure, and scheduling support for the chip.
- Aug '19 – **Career break** Inria, IRIF, and Université Paris Cité
Jan '23 Career break to pursue interest in mathematical logic. Audited courses in algebraic topology, and worked as an apprentice learning Rocq and type theory. Found a long-term collaborator, and began a line of research.
- Feb '15 – **Compiler Engineer** MathWorks, Natick, Massachusetts
Aug '17 Fixed the longest-standing bugs in the organization by carefully combing through x86 assembly. Worked with a pre-SSA IR to author and ship a LICM, and incremental-update algorithms for the program structure tree and dominator. Contributed to the in-house alias analysis. Implemented polyhedral loop optimizations using the *Integer Set Library*.

TALKS AND PUBLICATIONS

- Feb '26 **The very dependent recursive structure of iterated parametricity in indexed form**
with Hugo Herbelin
Pre-print: arXiv:2602.12689.
- Jun '25 **A parametricity-based formalization of semi-simplicial and semi-cubical sets**
with Hugo Herbelin
Mathematical Structures in Computer Science
Published: 10.1017/S096012952500009X. Pre-print: arXiv:2401.00512.
- Apr '25 **Making LoopAccessAnalysis more precise**
EuroLLVM, Berlin
On the limitations and the recent improvements to the analysis behind the vectorizer.
Program: LLVM Developers' Meeting. Recorded video: YouTube.

2014	Masters in Computer Engineering	Columbia University, New York
2012	Bachelors and Masters in Physics	Indian Institute of Technology, Kharagpur
2013 – 2014	Linux Landed 60 patches focused on improving perf tools.	
2010 – 2014	Git Landed 200 patches. Participated in Summer of Code 2010 and 2011, authoring the sequencer, enabling <code>git cherry-pick</code> and <code>git revert</code> to resume after conflicts. Authored <code>git rebase --autostash</code> and <code>remote.pushdefault</code> .	
2010	Subversion Authored <code>svnrump</code> , a tool to import and export history from a remote svn server.	